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Working Group XIII - Black Hole Accretion and Feedback

Working Group XIII - Blog

Untitled Post This is the first blog post for the Working Group XIII – Black Hole Accretion and Feedback.

Posted Oct 4, 2012, 1:00 AM by Ji-hoon Kim

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– Science Question: Effect of black hole feeding and feedback on the evolution of galaxies (isolated and cosmological) across participating codes, etc.

– Description: This working group will investigate the effect that including a black hole (BH) sub-grid model has on the overall dynamics of a galaxy across the suite of codes. In particular we will study the role that different modes and implementations of BH feedback have on galactic properties including (but not limited to) SFR, stellar morphology, M-sigma and M-Mbulge ratios, and relative strength between SF feedback and BH feedback across the lifetime of the simulation. The interplay between star formation and black hole growth will also be a priority. Since the simulations will be running anyway without BH sub-grid physics, the plan is that interested participants can simply re-run these simulations with SMBH feeding and feedback switched on, thus minimising extra work. The difference between running with and without an BH sub-grid model would be interesting for both the isolated discs, where the BH will start already at a typical SMBH mass and therefore exert a strong effect on the dynamics of the central region, and for the cosmological runs, where the BH would start as a seed and co-evolve with the galaxy as it forms. **[authored by Alexander Hobbs]**

Some suggestions:

(1) Regarding the Bondi accretion: We can use Bondi accretion ***for now***. I suggest we also compare alternative prescriptions for accretion.

(2) Regarding the feedback: I think coupling energy with neighboring particles/grid cells is dangerous, at least for SPH, as the results depend on the number of neighbors one choses. We can do it for now, but I suggest that we all use the same number of neighbors so we can have a meaningful comparison.

(3) I suggest we also compare different feedback prescriptions.

[edited by Amit Kashi]